Incidence Scheme

The universal family on a Hilbert scheme gives a scheme of pairs consisting of an embedded subscheme and a point on it. We study its two projections, first for all lines in the projective plane and then for the pencil of lines through a fixed point. The latter incidence scheme is the blow-up of the plane at that point. The same construction extends to projective space of arbitrary dimension.

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Chapter 1

Universal Families and Incidence Schemes

All schemes are over $\mathbb{C}$. Unless a test scheme is explicitly allowed to be more general, schemes are of finite type over $\mathbb{C}$. A curve in a Hilbert family means a one-dimensional closed subscheme; reducedness and irreducibility are not implicit in this terminology.

We use the *lines convention* for projectivization: if E is a locally free sheaf, then

$$\mathbb{P}(E) = \operatorname{Proj}\bigl(\operatorname{Sym} E^{\vee}\bigr).$$

Thus the fibre over a point parameterizes one-dimensional linear subspaces of the corresponding vector space. Hartshorne's convention in [Har77], II, §7, is $\operatorname{Proj}(\operatorname{Sym} E)$, which instead parameterizes one-dimensional quotients of E.

1.1 The Universal Family

1.1A The Hilbert scheme

Theorem 1.1.1 (Hilbert scheme)**.** *Let X be a projective scheme, let L be a fixed very ample line bundle on X, and let P be a numerical polynomial. There is a projective scheme*

$$H = \operatorname{Hilb}^{P,L}(X)$$

representing the contravariant functor

$$\operatorname{Hilb}_X^{P,L} : (\mathrm{Sch}/\mathbb{C})^{\mathrm{op}} \longrightarrow \mathrm{Sets},$$

which assigns to a scheme T the set of closed subschemes $Z \subset X \times T$ such that $Z \to T$ is flat and of finite presentation, and every geometric fibre has Hilbert polynomial P with respect to L. The morphism $Z \to T$ is projective, hence proper, because Z is closed in $X \times T$.

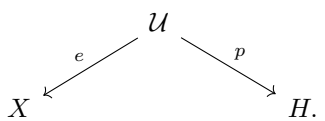
The action on a morphism $T' \to T$ is pullback:

$$Z \longmapsto Z \times_T T' \subset X \times T'.$$

The representing property gives a closed subscheme

$$\mathcal{U} \subset X \times H,$$

called the *universal family*, and two morphisms



Here p is the universal flat projective family, and e is the restriction of the other projection. For every Hilbert family $Z \subset X \times T$, there is a unique morphism $f : T \to H$ such that

$$Z = (\mathrm{id}_X \times f)^{-1}(\mathcal{U}) = \mathcal{U} \times_H T$$

as closed subschemes of $X \times T$. The equality is scheme-theoretic, not merely an equality of sets of closed points.

References. For existence of the Hilbert scheme, see [Gro61], Exposé 221. An explicit construction is given in [Nit05], §5, Theorem 5.1, applied to $E = \mathcal{O}_X$. For Hilbert polynomials, see [Har77], I, §7; for their behaviour in flat projective families, III, Theorem 9.9.

Definition 1.1.2 (Embedded subscheme)**.** An *embedded subscheme of* X is a pair (Z, i), where $i : Z \hookrightarrow X$ is a closed immersion. Two such pairs are identified if there is an isomorphism between their sources commuting with the closed immersions into X.

In particular, a Hilbert point remembers how the subscheme lies in X, not just its abstract isomorphism class. If $h \in H(\mathbb{C})$, the fibre

$$Z_h = p^{-1}(h)$$

is the corresponding embedded subscheme of X. Consequently, the closed points of the *universal family* are

$$\mathcal{U}(\mathbb{C}) = \{(x, h) \in X(\mathbb{C}) \times H(\mathbb{C}) : x \in Z_h\}. \tag{1.1.1}$$

This formula describes the closed points of $\mathcal{U}$. Its scheme structure is supplied by the universal family, and cannot in general be recovered from the set in (1.1.1).

1.1B Pulling back a family of curves

Definition 1.1.3 (Incidence scheme of a family)**.** Suppose that P has degree one, and let $f : B \to H$ be a morphism from an algebraic scheme B. The *incidence scheme* of the family selected by f is

$$\mathcal{U}_B = \mathcal{U} \times_H B \subset X \times B.$$

Its family projection and evaluation morphism are denoted by

$$p_B : \mathcal{U}_B \longrightarrow B, \qquad e_B : \mathcal{U}_B \longrightarrow X.$$

The construction fits into the diagram

$$\begin{array}{ccccc} \mathcal{U}_B & \xrightarrow{\quad} & \mathcal{U} & \xrightarrow{\ e\ } & X \\ {\scriptstyle p_B}\downarrow & & \downarrow{\scriptstyle p} & & \\ B & \xrightarrow[f]{} & H. & & \end{array} \qquad \text{with } e_B : \mathcal{U}_B \to X \text{ the composite}$$

The left square is Cartesian. Since both flatness and projectivity are preserved by base change, p_B is flat and projective. For $b \in B(\mathbb{C})$, write

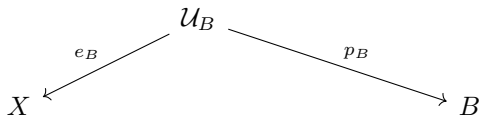
$$C_b = p_B^{-1}(b) \subset X.$$

Then

$$\mathcal{U}_B(\mathbb{C}) = \{(w, b) \in X(\mathbb{C}) \times B(\mathbb{C}) : w \in C_b\}. \tag{1.1.2}$$

Thus a point of the incidence scheme consists of a chosen member of the family together with a point on that member.

There are two ways to read the same pair (w, b):



w (the member C_b).

The map p_B asks which curve has been chosen. The map e_B asks where its distinguished point lies in X.

Remark 1.1.4 (The two kinds of fibres)**.** The fibre $p_B^{-1}(b)$ is the curve C_b itself. On the other hand, for a closed point $w \in X$, the scheme-theoretic fibre

$$e_B^{-1}(w) = \mathcal{U}_B \times_X \operatorname{Spec} \mathbb{C}$$

is naturally a closed subscheme of B; its closed points are the parameters b for which $w \in C_b$. Thus it parameterizes the members of the family which pass through w. If two different parameters determine the same curve, both parameters remain in this fibre.

The closed immersion into B in Remark 1.1.4 follows by base change from $\mathcal{U}_B \hookrightarrow X \times B$ along $\{w\} \times B \hookrightarrow X \times B$. In particular, the incidence condition also has a scheme structure, including any infinitesimal information carried by the family.

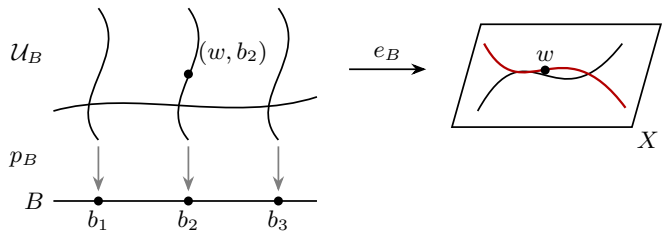


Figure 1.1: The family projection remembers the parameter; evaluation remembers the point in the target. The curves on the right are their images in X. The drawing is schematic.

Remark 1.1.5 (Drawing a flat family)**.** A picture generally shows only the supports of the fibres. For example, [Har77], III, §9, Figure 11, p. 260, depicts a flat family of subschemes of $\mathbb{P}^3$. For $a \neq 0$, its affine part is parameterized by

$$(x, y, z) = (t^2 - 1, t^3 - t, at).$$

The support of the special fibre at $a = 0$ is the nodal plane cubic $y^2 = x^2(x + 1)$, but the special fibre also has an embedded point at the node. This embedded structure is essential for flatness and is invisible in an ordinary drawing. Figure 1.2 illustrates the supports of several members of this family.

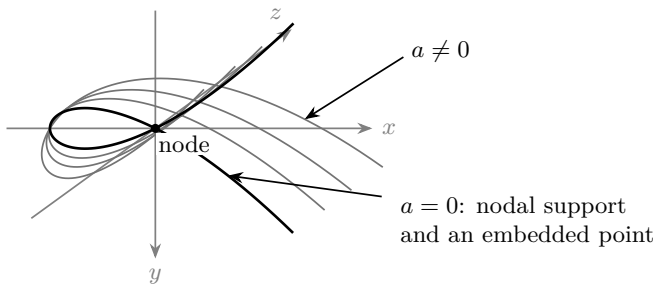


Figure 1.2: A spatial sketch inspired by [Har77], III, Figure 11. Only affine portions are drawn. At $a = 0$ the two branches meet; the embedded point is not represented by the support.

1.2 Point–Line Incidence in the Projective Plane

1.2A The dual plane

Let $V = \mathbb{C}^3$ and write $X = \mathbb{P}(V) = \mathbb{P}^2$, with homogeneous coordinates $[X : Y : Z]$. A point

$$[A : B : C] \in \mathbb{P}(V^\vee) = (\mathbb{P}^2)^\vee$$

defines the line

$$\ell_{[A:B:C]} = \{AX + BY + CZ = 0\} \subset \mathbb{P}^2.$$

Conversely, a line $\ell = \mathbb{P}(W)$, where $W \subset V$ has dimension two, determines the one-dimensional annihilator

$$W^\perp = \{\lambda \in V^\vee : \lambda|_W = 0\} \cong (V/W)^\vee.$$

Thus lines in $\mathbb{P}^2$ are parameterized by the dual projective plane. More precisely, the Hilbert scheme of lines, with Hilbert polynomial $P(m) = m + 1$ relative to $\mathcal{O}_{\mathbb{P}^2}(1)$, is $(\mathbb{P}^2)^\vee$.

Definition 1.2.1 (Point–line incidence scheme)**.** The *point–line incidence scheme* is the closed subscheme

$$I = \{AX + BY + CZ = 0\} \subset \mathbb{P}^2 \times (\mathbb{P}^2)^\vee. \tag{1.2.1}$$

We denote its projections by

$$p : I \longrightarrow (\mathbb{P}^2)^\vee, \qquad e : I \longrightarrow \mathbb{P}^2.$$

The polynomial in (1.2.1) is bihomogeneous of bidegree $(1, 1)$, so it defines a closed subscheme of the product. Under the Hilbert scheme identification above, I is the universal family of lines.

For a fixed $[A : B : C]$, the fibre of p is the line with that equation:

$$p^{-1}([A : B : C]) = \ell_{[A:B:C]} \cong \mathbb{P}^1.$$

For a fixed point $w = [X : Y : Z]$, the fibre of e is instead

$$e^{-1}(w) = \{[A : B : C] : AX + BY + CZ = 0\} \cong \mathbb{P}^1.$$

Indeed, evaluation at a nonzero representative of w is a nonzero linear functional on $V^\vee$, so its kernel has dimension two. This projective line in the dual plane is the *pencil of lines through* w.

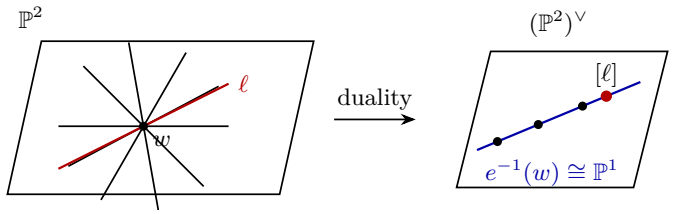


Figure 1.3: A pencil in the plane is a projective line in the dual plane. Each line through w gives one point of that projective line.

1.2B The two projective bundle structures

Proposition 1.2.2. *Both p and e are smooth $\mathbb{P}^1$-bundles. In particular, I is a smooth projective threefold.*

Proof. On $\mathbb{P}(V^\vee)$, the tautological line subbundle $\mathcal{O}(-1) \hookrightarrow V^\vee \otimes \mathcal{O}$ gives, by duality, a surjection

$$V \otimes \mathcal{O} \longrightarrow \mathcal{O}(1).$$

Let $\mathcal{K}$ be its kernel. Since the quotient is a line bundle, $\mathcal{K}$ is locally free of rank two. At a point $[\lambda] \in \mathbb{P}(V^\vee)$, its fibre is $\ker \lambda \subset V$. We claim that

$$I \cong \mathbb{P}(\mathcal{K})$$

over $\mathbb{P}(V^\vee)$. We verify this by an explicit calculation.

On the standard affine chart $U_C = \{C \neq 0\}$, put

$$a = A/C, \qquad b = B/C, \qquad R = \mathbb{C}[a, b].$$

The surjection becomes

$$R^3 \longrightarrow R, \qquad (f_1, f_2, f_3) \longmapsto af_1 + bf_2 + f_3.$$

Consequently,

$$\mathcal{K}(U_C) = \{(f_1, f_2, -af_1 - bf_2) : f_1, f_2 \in R\} \cong R^2.$$

Its dual consists of the functionals

$$\lambda_{g_1,g_2}(f_1, f_2, -af_1 - bf_2) = g_1 f_1 + g_2 f_2, \qquad g_1, g_2 \in R.$$

Restriction of functionals on R^3 induces the surjection

$$R^3 \longrightarrow \mathcal{K}^\vee(U_C), \qquad (h_1, h_2, h_3) \longmapsto \lambda_{h_1 - ah_3,\, h_2 - bh_3}.$$

Its kernel is the free rank-one submodule $R(a, b, 1)$. Hence

$$\mathcal{K}^\vee(U_C) \cong R^3/R(a, b, 1),$$

and, with the generators in degree one,

$$\operatorname{Sym}_R\big(\mathcal{K}^\vee(U_C)\big) \cong R[T_1, T_2, T_3]/(aT_1 + bT_2 + T_3). \tag{1.2.2}$$

Taking relative Proj gives

$$\begin{aligned}\mathbb{P}(\mathcal{K})|_{U_C} &= \operatorname{Proj}_R(R[T_1,T_2,T_3]/(aT_1+bT_2+T_3))\\ &= \{aT_1+bT_2+T_3=0\}\subset U_C\times\mathbb{P}^2.\end{aligned}$$

This is exactly $I|_{U_C}$. The same calculation on U_A and U_B gives the other local identifications. They agree on overlaps because each is induced by the inclusion $\mathcal{K}\hookrightarrow V\otimes\mathcal{O}$. Thus they glue to $I\cong\mathbb{P}(\mathcal{K})$.

Since $\mathcal{K}$ has rank two, this makes p a locally trivial $\mathbb{P}^1$-bundle, hence a smooth morphism. Interchanging V and $V^\vee$ gives the same conclusion for e. More explicitly, on $\mathbb{P}(V)$ one can take

$$\mathcal{K}' = \ker\bigl(V^\vee\otimes\mathcal{O}\longrightarrow\mathcal{O}(1)\bigr),$$

whose projectivization parameterizes linear forms vanishing on the chosen point. Then $I\cong\mathbb{P}(\mathcal{K}')$ over $\mathbb{P}(V)$.

The base $\mathbb{P}^2$ is smooth of dimension two, and the fibres have dimension one. Therefore I is smooth of dimension three. It is projective because it is closed in the product of two projective planes. □

The fact that the fibres are all isomorphic to $\mathbb{P}^1$ would not, by itself, establish the asserted bundle structures. The kernel calculation supplies the local trivializations.

References. For relative Proj and projective bundles, see [Har77], II, §7, especially Proposition 7.11; remember the difference in projectivization convention.

1.3 A Pencil of Lines and the Blow-up

1.3A Selecting the lines through a fixed point

Fix

$$z=[0:0:1]\in\mathbb{P}^2.$$

We parameterize directions through z by $[s:t]\in\mathbb{P}^1$. The corresponding line is

$$\ell_{[s:t]}=\{tX-sY=0\}.$$

Every line through z has this form. Indeed, the line $AX+BY+CZ=0$ passes through z exactly when $C=0$. Thus the pencil is selected by the morphism

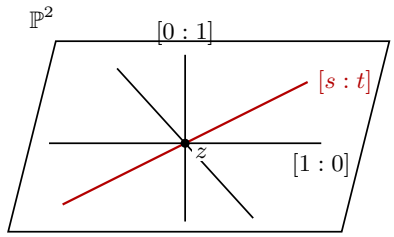
$$j:\mathbb{P}^1\longrightarrow(\mathbb{P}^2)^\vee,\qquad [s:t]\longmapsto[t:-s:0]. \tag{1.3.1}$$

Pulling back the universal incidence scheme gives

$$\begin{aligned}\mathcal{U}_z &= I\times_{(\mathbb{P}^2)^\vee}\mathbb{P}^1\\ &= \{tX-sY=0\}\subset\mathbb{P}^2\times\mathbb{P}^1.\end{aligned} \tag{1.3.2}$$

Its two projections fit into the diagram

$$\begin{array}{ccccc}\mathcal{U}_z & & \xrightarrow{\quad} & & I & \xrightarrow{\ e\ } & \mathbb{P}^2\\ & \overset{e_z}{\searrow} & & & & & \\ {\scriptstyle p_z}\downarrow & & & & \downarrow{\scriptstyle p} & & \\ \mathbb{P}^1 & & \xrightarrow[\ j\ ]{} & & (\mathbb{P}^2)^\vee. & & \end{array}$$



a direction $[s:t]$ determines the line $tX = sY$

Figure 1.4: The chosen pencil. In the affine chart $Z \neq 0$, the direction vector of $tX = sY$ is (s, t).

The morphism p_z is a smooth $\mathbb{P}^1$-bundle by base change from p, and $p_z^{-1}([s:t])$ is the line $\ell_{[s:t]}$. The evaluation morphism e_z behaves differently.

If $w = [X : Y : Z] \neq z$, there is exactly one line through z and w, with direction $[X : Y]$. In fact,

$$w \longmapsto (w, [X : Y]) \tag{1.3.3}$$

is an inverse to e_z away from z. Over z, every direction is allowed. Therefore

$$E = e_z^{-1}(z) = \{z\} \times \mathbb{P}^1 \subset \mathcal{U}_z.$$

The map e_z contracts E to z, whereas $p_z|_E : E \to \mathbb{P}^1$ is an isomorphism. Equivalently, E gives the section $[s:t] \mapsto (z, [s:t])$ of p_z.

1.3B Identifying the evaluation morphism

Proposition 1.3.1. *The evaluation morphism*

$$e_z : \mathcal{U}_z \longrightarrow \mathbb{P}^2$$

identifies $\mathcal{U}_z$ with $\operatorname{Bl}_z \mathbb{P}^2$. Under this identification, E is the exceptional divisor.

Proof. Work first on the affine chart $Z \neq 0$ containing z, and set

$$x = X/Z, \qquad y = Y/Z.$$

Then z becomes the origin, and the incidence equation becomes

$$xt = ys.$$

For $R = \mathbb{C}[x, y]$, the Rees algebra of the ideal $\mathfrak{m} = (x, y)$ has the graded presentation

$$\bigoplus_{d \geq 0} \mathfrak{m}^d u^d \cong R[S, T]/(xT - yS), \qquad S \longmapsto xu, \quad T \longmapsto yu.$$

Here u is an auxiliary variable and S, T have degree one. Thus

$$\begin{aligned} \operatorname{Bl}_{(0,0)} \mathbb{A}^2 &= \operatorname{Proj}_R \left(\bigoplus_{d \geq 0} \mathfrak{m}^d \right) \\ &= \{xt = ys\} \subset \mathbb{A}^2 \times \mathbb{P}^1. \end{aligned}$$

This is $e_z^{-1}(\{Z \neq 0\})$, with the same projection to $\mathbb{A}^2$.

The two standard charts make the identification concrete. On $s \neq 0$, put $v = t/s$. Then $y = xv$, and the map to the plane is

$$\mathbb{A}^2_{x,v} \longrightarrow \mathbb{A}^2_{x,y}, \qquad (x, v) \longmapsto (x, xv).$$

On $t \neq 0$, put $u = s/t$. Then $x = yu$, and the map is

$$\mathbb{A}^2_{u,y} \longrightarrow \mathbb{A}^2_{x,y}, \qquad (u, y) \longmapsto (yu, y).$$

The exceptional divisor is given by $x = 0$ in the first chart and by $y = 0$ in the second. These affine lines glue to $E \cong \mathbb{P}^1$.

Finally, away from z, both $\mathrm{Bl}_z \mathbb{P}^2 \to \mathbb{P}^2$ and e_z are isomorphisms. The local identification over $Z \neq 0$ agrees there with the inverse in (1.3.3). It therefore glues with the identification over $\mathbb{P}^2 \setminus \{z\}$, proving the global assertion. □

References. For the elementary incidence description of blowing up the origin, see [Har77], I, §4, the construction preceding Example 4.9.1, and Figure 3, p. 29. For the scheme-theoretic definition by the Rees algebra and its properties, see II, §7, especially Proposition 7.13.

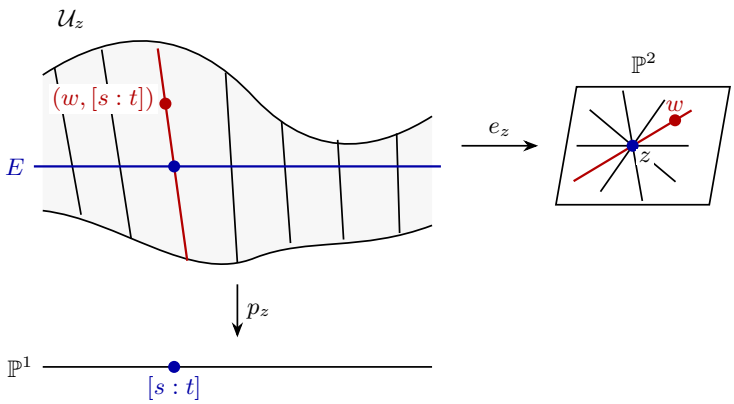


Figure 1.5: The incidence surface of the pencil. Each ruling is a fibre of p_z and maps isomorphically to the corresponding line through z. The section E is contracted by e_z. The ribbon is schematic; the fibres and E are complete projective lines. Compare [Har77], I, Figure 3.

Remark 1.3.2 (Flatness belongs to the family projection)**.** Although p_z is a smooth $\mathbb{P}^1$-bundle, e_z is not flat. Its fibres are a reduced point over $w \neq z$ and $\mathbb{P}^1$ over z. For example, restrict the relatively very ample line bundle $\mathcal{O}_{\mathbb{P}^2 \times \mathbb{P}^1}(0, 1)$ to $\mathcal{U}_z$. The fibre Hilbert polynomials for e_z are 1 away from z and $m + 1$ at z, which contradict local constancy in a flat projective family. See [Har77], III, Theorem 9.9.

The base-change square above concerns p_z and p. It gives no assertion that the other projection e_z is flat. This example distinguishes the two roles of an incidence scheme particularly clearly.

1.4 Lines Through a Point in Higher Dimension

The same picture extends to $\mathbb{P}^n$. Assume $n \geq 2$, write $V = \mathbb{C}^{n+1}$, and choose homogeneous coordinates

$$[X_1 : \cdots : X_n : X_{n+1}].$$

Fix the point $z = [0 : \cdots : 0 : 1]$. A line through z is determined by a direction

$$[s_1 : \cdots : s_n] \in \mathbb{P}^{n-1}.$$

In dimensions greater than two, the dual projective space parameterizes hyperplanes, rather than lines. Lines in $\mathbb{P}^n$ correspond to two-dimensional subspaces of V and are therefore parameterized by

$$G = \mathrm{Gr}(2, V).$$

1.4A The universal point–line incidence scheme

Let $\mathcal{S} \subset V \otimes \mathcal{O}_G$ be the tautological rank-two subbundle, whose fibre at $[W] \in G$ is W. The universal incidence scheme is

$$I = \mathbb{P}(\mathcal{S}) \subset \mathbb{P}(V) \times G.$$

Its closed points are the pairs $([v], [W])$ with $v \in W$. To write equations, choose a basis u, v of W, and let

$$p_{ij} = u_i v_j - u_j v_i, \qquad 1 \leq i < j \leq n+1,$$

be its Plücker coordinates. Membership of X in W is the condition

$$X \wedge (u \wedge v) = 0.$$

Thus, inside $\mathbb{P}^n \times G$, the incidence scheme is given by

$$X_i p_{jk} - X_j p_{ik} + X_k p_{ij} = 0, \qquad 1 \leq i < j < k \leq n+1. \tag{1.4.1}$$

The ambient Grassmannian already imposes the Plücker relations. These incidence equations are scheme-theoretic: locally where a Plücker coordinate is invertible, they express X as a linear combination of the two local generators of $\mathcal{S}$.

The selection morphism for lines through z is

$$j : \mathbb{P}^{n-1} \longrightarrow G, \qquad [s_1 : \cdots : s_n] \longmapsto \langle (s_1, \ldots, s_n, 0), (0, \ldots, 0, 1) \rangle .$$

For this choice of basis, $p_{i,n+1} = s_i$ for $1 \leq i \leq n$ and $p_{ij} = 0$ for $1 \leq i < j \leq n$. Pulling back (1.4.1), we obtain

$$\begin{aligned} \mathcal{U}_z &= I \times_G \mathbb{P}^{n-1} \\ &= \{ X_i s_j - X_j s_i = 0 : 1 \leq i < j \leq n \} \\ &\qquad \subset \mathbb{P}^n \times \mathbb{P}^{n-1}. \end{aligned} \tag{1.4.2}$$

The family projection $p_z : \mathcal{U}_z \to \mathbb{P}^{n-1}$ is again a smooth $\mathbb{P}^1$-bundle. The evaluation morphism is denoted by $e_z : \mathcal{U}_z \to \mathbb{P}^n$.

References. For Grassmannians as parameter schemes and as Hilbert schemes of linear subspaces, see [Nit05], §1, "Elementary Examples, Exercises," items (2)–(3), pp. 5–6, and the construction at the end of that section. Nitsure uses the quotient convention; two-dimensional subspaces of V correspond to rank-two quotients of $V^\vee$. The projective bundle description of incidence follows from the relative Proj construction used in Proposition 1.2.2.

1.4B Local coordinates for the blow-up

Proposition 1.4.1. *The evaluation morphism identifies $\mathcal{U}_z$ with $\mathrm{Bl}_z\,\mathbb{P}^n$. Its exceptional divisor is*

$$E = \{z\} \times \mathbb{P}^{n-1},$$

and $p_z|_E$ is an isomorphism onto the space of directions.

Proof. On the affine chart $X_{n+1} \neq 0$, put $x_i = X_i/X_{n+1}$. Then (1.4.2) becomes

$$x_i s_j = x_j s_i, \qquad 1 \leq i < j \leq n.$$

This is the standard incidence description of the blow-up of $\mathbb{A}^n$ at the origin. Indeed, on the direction chart $s_1 \neq 0$, put

$$v_j = s_j/s_1 \quad (2 \leq j \leq n), \qquad x = x_1.$$

The equations give $x_j = x v_j$ for $j \geq 2$, and all the remaining incidence equations follow from these. Hence this chart is affine n-space, and evaluation is

$$\begin{aligned} \mathbb{A}^n &\longrightarrow \mathbb{A}^n, \\ (x, v_2, \ldots, v_n) &\longmapsto (x, xv_2, \ldots, xv_n). \end{aligned} \tag{1.4.3}$$

These are exactly the standard charts of

$$\mathrm{Bl}_{(0,\ldots,0)}\,\mathbb{A}^n = \mathrm{Proj}_{\mathbb{C}[x_1,\ldots,x_n]}\left(\bigoplus_{d\geq 0}(x_1,\ldots,x_n)^d\right).$$

The analogous formulas on $s_i \neq 0$ give the other charts, with the usual transition maps s_j/s_i. Thus the affine descriptions glue. On the chart in (1.4.3), the exceptional divisor is $x = 0$; its direction coordinates are unrestricted. Globally over the origin it is $\mathbb{P}^{n-1}$.

Outside z, evaluation is an isomorphism with inverse

$$[X_1 : \cdots : X_n : X_{n+1}] \longmapsto \big([X_1 : \cdots : X_{n+1}], [X_1 : \cdots : X_n]\big).$$

The affine blow-up identification and this identification away from z agree on their overlap. They therefore glue to the required isomorphism over $\mathbb{P}^n$. The description of E and its projection to the direction space follows directly from (1.4.2). □

The construction can also be read as resolving the linear projection

$$\mathbb{P}^n \dashrightarrow \mathbb{P}^{n-1}, \qquad [X_1 : \cdots : X_n : X_{n+1}] \longmapsto [X_1 : \cdots : X_n],$$

whose unique point of indeterminacy is z. The incidence scheme records the direction which is missing at that point:

$$\begin{array}{ccc} \mathrm{Bl}_z\,\mathbb{P}^n & \xrightarrow{\;p_z\;} & \mathbb{P}^{n-1} \\ {\scriptstyle e_z}\big\downarrow & \nearrow & \\ \mathbb{P}^n & & \end{array}$$

For $n = 2$, this is precisely the pencil construction above. A fibre of p_z is a line with its direction fixed, whereas the exceptional fibre of e_z contains all directions at the chosen point.

References

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